

EXP 17

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import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

# Mobile phone dataset

data = {
    "Brand": ["Apple", "Samsung", "Redmi", "Vivo", "Apple", "Samsung",
              "Redmi", "Vivo", "Apple", "Samsung", "Redmi", "Vivo"],
    "RAM": [4, 6, 6, 8, 8, 8, 4, 6, 12, 12, 8, 8],
    "Storage": [64, 128, 128, 128, 256, 256, 64, 128, 256, 512, 128, 256],
    "Camera": [12, 50, 48, 50, 48, 50, 50, 64, 48, 108, 50, 64],
    "Battery": [3200, 5000, 5000, 4500, 3500, 5000, 5000, 4500, 4500, 5000, 5000, 5000],
    "Price": [50000, 30000, 15000, 20000, 70000, 45000,
              18000, 25000, 80000, 90000, 22000, 30000]
}

df = pd.DataFrame(data)

# Encode brand names
encoder = LabelEncoder()
df["Brand"] = encoder.fit_transform(df["Brand"])

# Input and output
X = df.drop("Price", axis=1)
y = df["Price"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
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X, y, test_size=0.2, random_state=42
)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

print("R2 Score:", round(r2_score(y_test, y_pred), 2))
print("MAE:", round(mean_absolute_error(y_test, y_pred), 2))

# Predict new mobile
brand = encoder.transform(["Apple"])[0]

new_phone = [[brand, 8, 256, 48, 4500]]

price = model.predict(new_phone)[0]

print("\nPredicted Mobile Price: ₹", round(price, 2))
```

OUTPUT:

R2 Score: 0.71

MAE: 13908.86

Predicted Mobile Price: ₹ 61247.7